



EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-500-1
	Date: 18-02-2021

EDMS 20-500-1

**TECHNICAL SPECIFICATION
FOR
HEAD END SYSTEM**

Issue: FEB-2021 / Rev- 1



EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-500-1
	Date: 18-02-2021

CONTENTS

<u>NO</u>	<u>Description</u>	<u>Page</u>
1	Head End System - General Requirements	3
2	Head End System - Meter Data Readings	5
3	Head End System - Metering Field Operations and Maintenance	6
4	Head End System – Reports	6
5	Head End System – Security	7
6	Head End System - Utility Operator User Interface	8
7	Head End System - Integrated GIS Tool	8
8	Head End System - MDMS Interface	9



EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-500-1
	Date: 18-02-2021

Head End System

1. Head End System - General Requirements:

The Head End System (HES) shall process 350 Million Messages in 2 hours for demand response (connect and disconnect relay operations) and Interval Data.

- 1.1. The Head End System shall process 140 Million Messages between itself and the Meter Data Management System (MDMS) in 6-8 hours
- 1.2. The Head End System shall act as the central collection point for the data within the AMI system. The HES collects data from all of the Data Concentrators and stores the gathered data in a secure database. Once data is stored and analyzed on the server, the data shall be available for display via an easy-to-use, web-based graphical interface.
- 1.3. The Supplier shall provide the hardware and software required to run the HES including any elements to monitor, backup and restore the system.
- 1.4. The Head End System shall manage and archive data for five years such that it can be accessed by any Utility computer or handheld devices, both locally and remotely via the web.
- 1.5. Mandatory - All data used within the solution shall be the sole property of the Distribution Company (DISCO), as cannot be used by, shared with or demonstrated to any third party. Any misuse of this data will be subject to the Egyptian Law.
- 1.6. The Head End System software must be web browser-based and shall have defined applications with standard interfaces to allow for existing and planned software applications.
- 1.7. The Head End System software must have flexible meter reading data output formats that are compatible with the DISCOs current billing application
- 1.8. The HES shall have the ability to pass along alerts and status codes to upstream systems for publication to the consumers as configured by the HES user interface.
- 1.9. Each meter-generated alert shall be accompanied by a duration the alert has been active that shall be stored and optionally sent out by the server.
- 1.10. For Managed Hosting Solutions, the Host Server shall be on an open system platform and support multiple operating systems, and shall work with any database system server



EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-500-1
	Date: 18-02-2021

software such as that provided by Microsoft, Oracle, etc., and managed by the DISCO Personnel in cooperation with Vender Support.

- 1.11. For Perpetual License Solutions, the Supplier shall provide to the DISCO a database software server license.
- 1.12. The Host Software shall communicate with the Data Concentrator / WAN Operated meters via the open, international communication protocol, Device Language Message Specification (DLMS) / Companion Specification for Energy Metering (COSEM). Alternative solutions for the communication protocol shall be considered on merit.
- 1.13. The DLMS/COSEM profile(s) (data structure) will be defined, programmed, and tested during the proof-of-concept and pilot deployments. If alternative solutions have been provided in accordance with 3.1.13, the Supplier shall define, program, and test the profiles for those solutions. These definitions, programming, and testing shall be at no additional cost to the Utility.
- 1.14. Mandatory - The Head End System software shall have a commercially and practically proven track record, with a reference list of multiple deployed AMI projects of significantly size (at least 1 Million meters/endpoints) working for at least 2 years, accompanied by satisfaction letters from the project owners. At least 3 letters from different utilities are required.
- 1.15. The Head End System shall have a Certificate of Operation with different electricity meter/ Data Concentrators Suppliers (at least 3 meter Suppliers); this certificate shall be addressed from the successful project owners.
- 1.16. The Head End System should have a Certificate of successful integration with at least two reputable MDMS software solutions; this certificate shall be addressed from the successful project owners.
- 1.17. Mandatory - The bidder shall provide standard Multi Speak interfaces for integration to a future Meter Data Management System (MDMS), CRM, Outage Management System (OMS), and a Prepayment System (PPS) from the Head End System The successful bidder will also be able to support additional Multi Speak interfaces to additional systems in the future.
- 1.18. The successful bidder shall be able to submit their head end system (HES) to the Multi Speak testing / Bodies for certification during the evaluation period and pass the certification tests for the identified interfaces.
- 1.19. The HES software shall be in English and Arabic Language interface, with associated applications elements, menus, online help file, printed documentation, and training materials.



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 20-500-1

Date: 18-02-2021

2. Head End System - Meter Data Readings:

2.1. Head End System shall provide on-demand and scheduled reading of Smart Meters

- 2.1.1. Date and time
- 2.1.2. Energy reading profile data
- 2.1.3. Event log
- 2.1.4. Firmware version
- 2.1.5. Serial Number
- 2.1.6. Type
- 2.1.7. Other data points as configured by the DISCO (e.g., voltage, demand, status flags, etc.)

2.2. Head End System shall support two reading mechanisms, either pull or push operation, or a combination between them.

2.2.1. Pull operation means that the reading process will be initiated and controlled by the HES.

2.2.2. Push Operation means that a Data Concentrator / WAN Operated Smart Meter sends its data proactively to the Head End System, based on the following conditions:

2.2.2.1. Predefined Schedule, which previously configured in the Data Concentrator / WAN Operated Smart Meter.

2.2.2.2. Vital information (Data Concentrator / WAN Operated Smart Meter Alarms, Smart Meters newly registered).

2.2.3. The Reading Process shall be done on a batch of meters, or for a larger collection of Meters according to predefined Reading Process and Procedures

2.2.3.1. In case of non-complete reading data, the Head End System shall perform the reading process on the failed devices several times with different predefined interval time.

2.2.4. Head End System shall provide a flexible reading schedule that enables repetitions and start delays

2.2.5. Head End System shall provide a statistical analysis of the quality of communication performance.

2.2.6. Head End System shall provide a minimum set of System performance counters

2.2.6.1. Number of executed tasks

2.2.6.2. Number of failed tasks

2.2.6.3. Number of successful tasks

2.2.6.4. Maximum tasks performed per hour

2.2.6.5. Average execution time for scheduled reading tasks

2.2.7. Head End System shall provide an estimated reading performance before performing the Reading process, in order to facilitate decision making by the Utility operator.



EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-500-1
	Date: 18-02-2021

3. Head End System - Metering Field Operations and Maintenance:

3.1. Head End System shall support the processes and procedures of installation, replacement, and removal

3.1.1. For electricity meters

3.1.2. For Data Concentrators

3.2. Head End System shall provide static and dynamic grouping of devices:

3.2.1. Placing devices in static groups according to geographic regions, transformers, substations, etc.

3.2.2. Dynamic groups based on defined selection criteria.

3.2.3. These groups shall be automatically updated after installation, replacement, and removal devices.

3.3. Head End System shall provide a mechanism batch import of devices using device templates

3.3.1. These templates shall be verified and approved by the Agency.

3.4. The Head End System shall detect the newly installed meter / Data Concentrator, and label its Status with "Installed"

3.5. The Head End System shall configure the newly installed meter / Data Concentrators with the appropriate parameters

3.5.1. Tariff Scheme

3.5.2. Date and Time

3.5.3. Load Limit

3.5.4. Others as defined by the DISCO (e.g., encryption keys, demand limits, etc.)

4. Head End System – Reports:

4.1. The Host Software shall be used to generate reports; view demand graphs, determine usage patterns and enforce energy restrictions.

4.2. The reports shall be available in graphical and tabular views for reading and consumption for various intervals.

4.3. The graphs shall be available to show high/low temperatures for each day and rainfall on a daily basis.



EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-500-1
	Date: 18-02-2021

4.4. The reporting feature shall support multiple report generation/export formats, minimally

4.4.1.Extensible Markup Language (XML)

4.4.2.Comma-separated Values (CSV)

4.4.3.Microsoft Excel

4.4.4.Microsoft Word

4.4.5.Adobe PDF

5. Head End System – Security:

5.1. Head End System shall provide secured communication internally (among its components) and externally (to metering devices and upper-level systems).

5.2. Head End System shall support user and group security rights and polices for authentication, which based on the rights it grants access to its database and features:

5.2.1.Right to access Specific Data

5.2.2.Right to execute operations {reading, disconnection, reconnection, etc.}

5.2.3.Manage Devices.

5.2.4.Add/Remove/Edit User Accounts

5.3. The Head End System shall provide a secure remote access mechanism, log in and password.

5.4. The Head End System Software shall use a method for encryption/decryption of data exchanged between the Data Concentrator / WAN Operated Smart Meters and the Head End System.

5.4.1.At least ECC192 method shall be used in all information exchanges in public network as well as between concentrator and Head End System directly.

5.4.2.The security method and hardware should be described in details by the Bidder.

5.4.3.The HES shall be capable of supporting additional encryption methods over the life cycle of the system – it is expected that by the time this system is retired 256 bit keys will be required.

5.4.4.The HES shall be capable of managing encryption key rotation for the meters it communicates with.

5.4.5.The HES shall provide role-based access control.

5.5. The Bidder shall prove following items:

5.5.1. Security of method

5.5.2.Utility's operator ability to change security key remotely

5.5.3.Confidentiality of remote key changing

5.5.4.All input and output information decrypted and encrypted



EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-500-1
	Date: 18-02-2021

6. Head End System - Utility Operator User Interface:

- 6.1. The Head End System solution shall provide secure web-based user interface to the Utility personnel in order to facilitate their working procedures.
- 6.2. The User Interface shall permit the sending of alert outages, tampering, out-of- bounds system operating parameters to appropriate utility personnel via cell phone, pager, or e-mail.
- 6.3. The User Interface shall allow the utility to correlate consumption with meteorological data available on the user interface.
- 6.4. The Consumer data shall be available via a user-friendly interface that will allow for analysis, as well as bill generation.
- 6.5. The HES will support role based security for users
- 6.6. The HES will support a minimum of Arabic and English languages in the interface. Switching between languages shall be performed by setting parameters in the user login script.
- 6.7. Setting up user accounts will be done via a user-friendly web-linked interface by the properly authorized administrator-role user.

7. Head End System - Integrated GIS Tool:

- 7.1. The Host Software shall include a graphical information system (GIS) tool for network routing from Data Concentrators to AMI meters.
- 7.2. The GIS tool shall have icons that permit viewing system health of the star/mesh system components.
- 7.3. Routing distances from Data Concentrators to meters shall be part of the GIS tool.
- 7.4. The GIS tool shall allow monitoring of performance and loading on the utility's Infrastructure with the ability to display this information on GIS maps.



EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-500-1
	Date: 18-02-2021

- 7.5. The GIS tool shall allow monitoring of electric meter outage/restoration detection for electric meters.

8. Head End System - MDMS Interface

- 8.1. Head End System should provide multiple web service methods for being integrated with any MDMS and other Systems. The HES shall comply the Multi Speak protocol /Standard interface for messages and methods.
- 8.2. The awarded bidder shall have the responsibility to integrate with the System Integrator (MDM, Billing, Prepaid, Vending, etc.), that will be awarded in the subsequent RFP.
- 8.3. All interfaces from the HES to the MDMS systems shall conform to Multi Speak interface standards.
- 8.4. The Interface shall minimally provide a functionality level of:
- 8.4.1. Metering Devices Management (e.g. Connection/disconnections, Load Control of AMI meters)
 - 8.4.2. Triggering scheduled readings of metering data.
 - 8.4.3. Triggering on-demand readings of metering data
 - 8.4.4. Deploying Configuration files to meters and Data Concentrators (Tariff Schemes, Load Limits, etc.)
 - 8.4.5. Provide operational status notifications to the MDMS (e.g., meter reading status working / finished)
 - 8.4.6. Provide Push Notification alarms or events to MDMS.
 - 8.4.7. Vending (Postpaid / Prepaid) Related operation data.